

Monetary Economics and Policy

Pierpaolo Benigno

Chapter 7: Inflation Targeting as
an Optimal Policy



Prepared by Marcos Consuegra

What We Will Learn

- ▶ How the New Keynesian benchmark provides a theoretical foundation for **flexible inflation targeting** as an optimal policy strategy.
- ▶ How to derive the optimal targeting rule under **commitment** (specifically, the **timeless perspective**).
- ▶ How to distinguish optimal commitment from **discretion** and simple **interest-rate rules**, and how determinacy leads to the **Taylor principle**.
- ▶ How optimal policy under commitment introduces **history dependence** and **price-level reversion** to stabilize expectations.
- ▶ How to **rationalize and assess** the optimal numerical inflation target, and why many models imply a positive target in the **1–3% range**.

Outline

- 1 Introduction
- 2 Outline of the Results
- 3 Optimal policy under commitment
- 4 Optimal policy under commitment and markup shocks
- 5 Optimal Policy under Discretion
- 6 Optimal Inflation Target
- 7 Main Takeaways

- 1 Introduction
- 2 Outline of the Results
- 3 Optimal policy under commitment
- 4 Optimal policy under commitment and markup shocks
- 5 Optimal Policy under Discretion
- 6 Optimal Inflation Target
- 7 Main Takeaways

Inflation Targeting: Context and Theory

Historical Context

- ▶ Pioneered by New Zealand (1990), followed by Canada, UK, Sweden, and Finland.
- ▶ US Federal Reserve formally adopted a 2% target in 2012.
- ▶ *Key Features*: Public announcement of a numerical target (typically 1–3%) combined with central bank accountability.

Theoretical Rationalization

- ▶ Practical implementation of inflation targeting strategies preceded the development of suitable academic frameworks.
- ▶ The New Keynesian benchmark provides the **microfoundations** that rationalize inflation targeting as the optimal policy regime.
- ▶ This approach proves superior to simple instrument rules or strict adherence to the Wicksellian natural rate (r^*).

Optimal Policy: The Welfare Criterion

Microfounded Objective

- ▶ Policy evaluation is based on maximizing the household's expected utility.
- ▶ Since equilibrium conditions are log-linearized (first-order), a correct ranking of alternative policies requires a **second-order approximation** of utility.

The Central Bank's Welfare-Based Loss Function

$$L_{t_0}^{CB} = \mathbb{E}_{t_0} \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[\frac{1}{2} (\hat{Y}_t - \hat{Y}_t^e)^2 + \frac{1}{2} \frac{\theta}{\kappa} (\pi_t - \pi)^2 \right] \quad (7.1)$$

Definitions

- ▶ $\hat{Y}_t$: log deviation of output from steady state; $\hat{Y}_t^e$: *efficient* level of output.
- ▶ π_t : inflation rate; π : inflation target.
- ▶ $\beta \in (0, 1)$: households' discount factor; $\theta > 0$: elasticity of substitution across differentiated goods; $\kappa > 0$: slope of the New Keynesian Phillips curve.

The Policy Problem: Objectives and Constraints

Welfare-Theoretic Objective

- ▶ **Inflation stabilization:** Deviations $(\pi_t - \pi)$ create *price dispersion*, implying an inefficient allocation of demand across otherwise identical goods.
- ▶ **Real activity:** The welfare-relevant output gap is measured relative to the *efficient* level, $(\hat{Y}_t - \hat{Y}_t^e)$, and deviations from efficiency are costly.
- ▶ **Relative weight:** The welfare-based loss function pins down the optimal weight on inflation stabilization versus output-gap stabilization (via θ/κ).
- ▶ The loss is minimized when $\hat{Y}_t = \hat{Y}_t^e$ and $\pi_t = \pi$ *simultaneously*.

AS Constraint (New Keynesian Phillips Curve)

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta \mathbb{E}_t(\pi_{t+1} - \pi), \quad (7.2)$$

Nature of the Trade-off and Optimal Strategy

Shocks and Trade-offs

- ▶ Following productivity or public spending shocks (**efficient shocks**), the natural rate $\hat{Y}_t^n$ coincides with the efficient level $\hat{Y}_t^e$.
- ▶ Optimal policy stabilizes inflation at the target and simultaneously closes the output gap.
- ▶ A trade-off arises *only* with **markup shocks**, which create a wedge between the natural and efficient levels of output.

Characterization of Optimal Policy

- ▶ **Flexible Inflation Targeting** balances deviations from price stability with corresponding variations in output gap growth.
- ▶ **Commitment** implies **history dependence**: deviations of prices from the target trend are gradually undone to revert prices to the initial trend.

- 1 Introduction
- 2 Outline of the Results**
- 3 Optimal policy under commitment
- 4 Optimal policy under commitment and markup shocks
- 5 Optimal Policy under Discretion
- 6 Optimal Inflation Target
- 7 Main Takeaways

Outline of the Results

- ▶ Under **commitment** (the timeless perspective), the optimal policy yields targeting rules that rationalize **flexible inflation targeting**.
- ▶ **Discretion** and simple interest-rate rules are proven **sub-optimal** relative to the commitment benchmark.
- ▶ The superiority of commitment stems from **policy inertia** (history dependence) and the property of **price-level reversion**.
- ▶ Analysis of the optimal **numerical value** for the inflation target π .

- 1 Introduction
- 2 Outline of the Results
- 3 Optimal policy under commitment**
- 4 Optimal policy under commitment and markup shocks
- 5 Optimal Policy under Discretion
- 6 Optimal Inflation Target
- 7 Main Takeaways

Defining Optimal Policy under Commitment

- ▶ **Commitment:** The policymaker chooses *once and for all* a state-contingent path for the relevant macroeconomic variables and commits to implementing that plan in all future contingencies.
- ▶ **Not a Strict Rule:** Commitment does not imply adherence to a rigid instrument rule that is valid irrespective of the state of the economy. Instead, it consists of announcing and following an optimal, fully state-contingent plan.
- ▶ **Timeless Perspective:** A stronger form of commitment in which, even at time t_0 , the policymaker internalizes constraints on variables at t_0 that entered private-sector expectations formed in period $t_0 - 1$, thereby fulfilling past promises.
- ▶ **Discretion:** The policymaker re-optimizes each period, taking the current state of the economy as given, and is not bound by past plans or previously formed expectations.

Optimal Policy under Commitment: Timeless Perspective

- ▶ The policymaker minimizes the welfare-based loss function (7.1) subject to the sequence of AS constraints (7.2), adopting a **timeless perspective**.

$$\mathcal{L}_{t_0} = \mathbb{E}_{t_0} \left\{ \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[\frac{1}{2} (\hat{Y}_t - \hat{Y}_t^e)^2 + \frac{1}{2} \frac{\theta}{\kappa} (\pi_t - \pi)^2 + \lambda_t \left((\pi_t - \pi) - \kappa(\hat{Y}_t - \hat{Y}_t^n) - \beta(\pi_{t+1} - \pi) \right) \right] \underbrace{- \lambda_{t_0-1}(\pi_{t_0} - \pi)}_{\text{Timeless perspective constraint}} \right\}.$$

- ▶ The timeless perspective introduces an additional constraint on inflation at time t_0 , ensuring that expectations formed at time $t_0 - 1$ are fulfilled (promise keeping).
- ▶ **Note:** The AD equation is *not* a binding constraint; it determines the policy rate residually, provided the zero lower bound does not bind (see Chapter 9).

Optimality Conditions under Commitment: Timeless Perspective

- ▶ The first-order conditions with respect to inflation π_t and output $\hat{Y}_t$ are, for **each** $t \geq t_0$:

First-Order Conditions

$$\theta(\pi_t - \pi) + \kappa\lambda_t - \kappa\lambda_{t-1} = 0,$$

$$(\hat{Y}_t - \hat{Y}_t^e) - \kappa\lambda_t = 0.$$

Implications of the Timeless Perspective

- ▶ The inclusion of the initial constraint $\lambda_{t_0-1}(\pi_{t_0} - \pi)$ ensures that the first-order conditions have the same form in every period, including at t_0 .
- ▶ This renders the optimization problem **stationary** and the resulting policy **time consistent**.

Standard Commitment: Time Inconsistency Problem

Standard Commitment

- ▶ Under standard commitment at time t_0 , the policymaker maximizes welfare subject to the sequence of AS constraints involving *future* expectations (e.g. $E_t(\pi_{t+1} - \pi)$ for all $t \geq t_0$).
- ▶ Crucially, the optimization problem does *not* include an additional constraint on expectations formed at time $t_0 - 1$ concerning variables realized at time t_0 .

The Mechanism of Time Inconsistency

- ▶ **Re-optimization at $T > t_0$:** If the policymaker re-optimizes at a future date T , expectations formed at $T - 1$ regarding variables at time T do not appear as constraints in the optimization problem.
- ▶ **Result:** The set of binding constraints changes over time, so the optimal plan derived at time T generally differs from the plan announced at time t_0 .
- ▶ This generates an *ex-post* incentive to deviate from the *ex-ante* optimal plan, leading to time inconsistency.

Commitment from a Timeless Perspective

Credibility and Expectations

- ▶ The *timeless perspective* modifies the optimization problem at time t_0 by imposing an additional state-contingent constraint, $\lambda_{t_0-1}(\pi_{t_0} - \pi)$, which reflects expectations formed in period $t_0 - 1$.
- ▶ By internalizing past expectations, this constraint eliminates the **time-inconsistency problem** and the incentive to re-optimize in future periods.
- ▶ As a result, the optimization problem is **stationary**: the same set of constraints applies at all dates, and the resulting policy functions are time consistent.

Optimal Policy: The Targeting Rule

Derivation

- ▶ Combining the first-order conditions eliminates the Lagrange multipliers, yielding the **optimal targeting rule** under the timeless perspective:

$$(\pi_t - \pi) + \theta^{-1}(\Delta \hat{Y}_t - \Delta \hat{Y}_t^e) = 0. \quad (7.3)$$

Flexible Inflation Targeting

- ▶ (7.3) rationalizes a framework broader than strict inflation targeting by explicitly weighing real economic activity alongside price stability.
- ▶ Positive inflation deviations are allowed only if output growth falls below the efficient rate (i.e., $\Delta \hat{Y}_t < \Delta \hat{Y}_t^e$).

Elements of Flexible Inflation Targeting

Three Essential Elements

- ▶ The flexible inflation target includes both **inflation** (π_t) and **output growth** ($\Delta\hat{Y}_t$).
- ▶ **Fixed** inflation target π and the **time-varying** efficient growth trend $\Delta\hat{Y}_t^e$.
- ▶ The structural parameter θ dictates the optimal **relative weight** between the realizations of the variables and their targets.

Comparison with other Rules

- ▶ Unlike simple interest-rate rules or tracking the natural rate (r^*), this approach defines a **robust optimality condition**; the policy rate is adjusted endogenously to ensure the flexible inflation target holds regardless of the specific shock hitting the economy.

Equivalence with Price-Level Targeting

- The optimal targeting rule (7.3) is also equivalent, under commitment, to a **price-level targeting** criterion in which the central bank targets a composite reference price $\tilde{p}_t$ that combines the nominal price level and the real efficiency gap:

$$\tilde{p}_t = p_t + \theta^{-1}(\hat{Y}_t - \hat{Y}_t^e).$$

- The policymaker commits to keeping $\tilde{p}_t$ on a deterministic target path $\tilde{p}_t^*$ that grows at the inflation target rate π :

$$\tilde{p}_t^* = \tilde{p}_{t-1}^* + \pi.$$

Implications of the Price-Level Equivalence

Equivalence and Reversion

- ▶ The price-level targeting rule $\tilde{p}_t = \tilde{p}_t^*$ is **analytically equivalent** to the flexible inflation-targeting rule derived previously.
- ▶ Optimal policy requires the reference price to track the trending target, implying that past deviations must be offset (**history dependence**).
- ▶ **Note:** If the **ZLB binds** the equivalence between price-level targeting and inflation targeting breaks down, as shown in Chapter 9 and optimal policy is characterized by a price-level target.

Special Case: Nominal GDP Targeting

- ▶ If the structural parameter $\theta = 1$, the price-level rule simplifies to a **Nominal GDP targeting** rule.
- ▶ Nominal GDP ($p_t + y_t$) targets a reference path $\tilde{y}_t$:

$$p_t + y_t = \tilde{y}_t.$$

- ▶ The target $\tilde{y}_t$ evolves according to the inflation target and efficient output growth:

$$\tilde{y}_t = \tilde{y}_{t-1} + \pi + \Delta \hat{Y}_t^e.$$

- ▶ Nominal GDP targeting is nested as a specific parameterization of the optimal targeting rule.

Optimal Stabilization and Shock Types

Efficient Shocks (Productivity, Public Spending)

- ▶ The AS equation implies no trade-off between stabilizing inflation and closing the output gap relative to the natural level.
- ▶ Complete inflation stabilization is optimal, as the natural rate moves in tandem with the efficient rate.

Inefficient Shocks (Markup)

- ▶ These shocks generate a meaningful trade-off, which the targeting rule optimally manages.
- ▶ The policymaker tolerates inflation above target insofar as it is accompanied by a contraction of output growth relative to the efficient rate.

The IT Equation

- ▶ We provide a graphical representation of the optimal policy problem in the spirit of the AS–AD framework, augmented with the **IT curve**.
- ▶ Assuming $\pi = 0$ and that the economy is initially on the target path, the targeting rule (7.3) implies a negative relationship between the price level and output:

IT Equation

$$(p_t - p^*) + \theta^{-1}(\hat{Y}_t - \hat{Y}_t^e) = 0, \quad (7.4)$$

- ▶ The slope of the IT curve is $-1/\theta$.
- ▶ The IT curve is flatter than the AD curve whenever $\theta > \sigma$, which is the empirically relevant case.

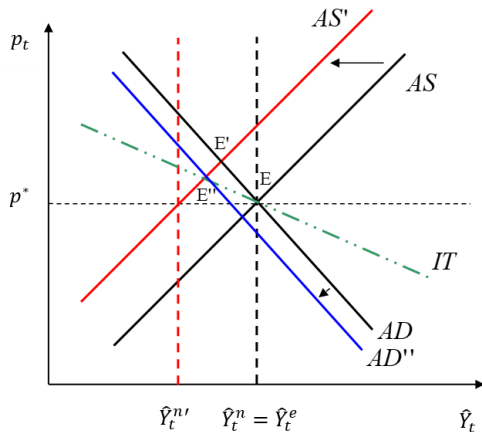
Optimal Response to a Markup Shock

Scenario

- ▶ A temporary **increase in the markup** ($\hat{\mu}_t \uparrow$): **AS shifts upward to AS'** . The *IT* locus remains fixed, as the shock is inefficient and $\hat{Y}_t^e$ is unchanged.

Equilibrium Analysis

- ▶ **Equilibrium moves from E to (E')** : Intersection of AS' and the initial AD . The economy exhibits stagflation.
- ▶ **Optimal Equilibrium (E'')**: Intersection of AS' and the *IT* curve. Monetary policy should raise the policy rates, contract the economy and move AD to AD' .



Policy Guidance: IT vs. Wicksellian Natural Rate

Inflation-Targeting Guidance

- ▶ IT gives the welfare-optimal combination of prices and output.
- ▶ Policy rate should adjust to shift AD along the AS curve until the economy reaches the locus implied by IT .

Wicksellian Contrast

- ▶ Wicksellian rule follows the frictionless real rate r_t^e .
- ▶ r_t^e varies only with efficient shocks $\rightarrow$ unchanged under markup shocks.
- ▶ Following the guidance of r_t^e when there are mark-up shocks delivers a non-optimal allocation.

- 1 Introduction
- 2 Outline of the Results
- 3 Optimal policy under commitment
- 4 Optimal policy under commitment and markup shocks**
- 5 Optimal Policy under Discretion
- 6 Optimal Inflation Target
- 7 Main Takeaways

Optimal Policy: Dynamic Representation

- It is convenient to rewrite the system in terms of the gap between output and its *efficient* level:

$$x_t \equiv \hat{Y}_t - \hat{Y}_t^e.$$

- **Rewriting the IS Equation (AD)**

$$x_t = E_t x_{t+1} - \sigma (\hat{i}_t - E_t(\pi_{t+1} - \pi) - r_t^e). \quad (7.5)$$

- **Rewriting the Phillips Curve (AS)**

$$\pi_t - \pi = \kappa x_t + v_t + \beta E_t(\pi_{t+1} - \pi). \quad (7.6)$$

Optimal Policy: Dynamic Representation

The Cost-Push Shock (v_t)

$$v_t = \kappa(\hat{Y}_t^e - \hat{Y}_t^n) = \frac{\kappa}{\sigma^{-1} + \eta} \hat{\mu}_t.$$

The Frictionless Real Rate (r_t^e)

- Defined as the rate consistent with flexible prices and efficient output. It summarizes disturbances from productivity, spending, and preferences:

$$r_t^e \equiv E_t \left\{ \frac{1 + \eta}{1 + \sigma\eta} (\hat{A}_{t+1} - \hat{A}_t) - \frac{\eta}{1 + \sigma\eta} (\hat{G}_{t+1} - \hat{G}_t) - (\hat{\xi}_{t+1} - \hat{\xi}_t) \right\}.$$

Optimal Policy: Dynamic Representation

- ▶ Combining the AS equation (7.6) with the targeting rule (7.3) allows us to eliminate inflation.
- ▶ This yields a second-order stochastic difference equation for the output gap x_t :

$$E_t x_{t+1} - \left(1 + \frac{1}{\beta} + \frac{\kappa\theta}{\beta}\right) x_t + \frac{1}{\beta} x_{t-1} = \frac{\theta}{\beta} v_t. \quad (7.7)$$

- ▶ The dynamic properties of the optimal allocation depend on the roots of the associated characteristic polynomial $P(\gamma)$:

$$P(\gamma) = \gamma^2 - \left(1 + \frac{1}{\beta} + \frac{\kappa\theta}{\beta}\right) \gamma + \frac{1}{\beta}. \quad (7.8)$$

Optimal Policy: Determinacy and Stability

Conditions for Determinacy (Unique and Stable Solution)

- ▶ **Uniqueness:** Essential for effective policy control. Without uniqueness, multiple equilibrium paths would be consistent with the same policy, undermining the ability to control inflation and output.
- ▶ **Stability:** Required for the validity of the log-linear approximation, which is meaningful only for solutions that remain close to the steady state.

Blanchard–Kahn Conditions

- ▶ A linear rational expectations system has a unique and stable solution if the number of eigenvalues inside the unit circle equals the number of predetermined variables.
- ▶ The difference equation (7.7) contains one predetermined variable, x_{t-1} .
- ▶ Therefore, the characteristic polynomial must have exactly **one eigenvalue inside the unit circle** (stable root) and **one eigenvalue outside** (unstable root).

Conditions for Determinacy

Determinacy and Stability

Properties of the Polynomial $P(\gamma)$

- ▶ We evaluate $P(\gamma)$ at critical points to locate the roots:

$$P(0) = \frac{1}{\beta} > 0, \quad P(1) = -\frac{\kappa\theta}{\beta} < 0, \quad P(\infty) = \infty$$

Location of Eigenvalues

- ▶ Since $P(\gamma)$ is continuous and changes sign between 0 and 1, the first root satisfies $0 < \gamma_1 < 1$.
- ▶ The product of roots is $\gamma_1\gamma_2 = 1/\beta$. Given $\gamma_1 < 1$ and $\beta < 1$, the second root must satisfy $\gamma_2 > 1$.
- ▶ **Conclusion:** The eigenvalues satisfy $0 < \gamma_1 < 1 < \gamma_2$. The determinacy condition is met.

Solving the Difference Equation: Step 1

Defining the Lag Operator

- ▶ Let us define the lag operator (L) with properties $L \cdot x_t = x_{t-1}$ and $L^{-1} \cdot x_t = E_t x_{t+1}$.
- ▶ More generally, $L^k \cdot x_t = x_{t-k}$ and $L^{-k} \cdot x_t = E_t x_{t+k}$.

Rewriting the Equation

- ▶ We can then write equation (7.7) as:

$$\left[L^{-1} - \left(1 + \frac{1}{\beta} + \frac{\kappa\theta}{\beta} \right) + \frac{1}{\beta}L \right] x_t = \frac{\theta}{\beta} v_t.$$

Substituting Roots

- ▶ Note that $\gamma_1 + \gamma_2 = \left(1 + \frac{1}{\beta} + \frac{\kappa\theta}{\beta} \right)$ and $\gamma_1\gamma_2 = \frac{1}{\beta}$.
- ▶ Therefore, using the sum and product of roots:

$$\left[L^{-1} - (\gamma_1 + \gamma_2) + \gamma_1\gamma_2 L \right] x_t = \theta\gamma_1\gamma_2 v_t.$$

Solving the Difference Equation: Step 2

Factorization

- ▶ We can factor the previous expression into stable and unstable components:

$$(1 - \gamma_1 L)(L^{-1} - \gamma_2)x_t = \theta\gamma_1\gamma_2 v_t.$$

Simplification

- ▶ We can further simplify the above expression to:

$$(1 - \gamma_1 L)(1 - \gamma_2^{-1} L^{-1})x_t = -\theta\gamma_1 v_t.$$

Forward solution for the unstable block

- ▶ It follows that:

$$\begin{aligned}(1 - \gamma_1 L)x_t &= -\frac{\theta\gamma_1}{(1 - \gamma_2^{-1} L^{-1})}v_t \\ &= -\theta\gamma_1 E_t \sum_{j=0}^{\infty} \gamma_2^{-j} v_{t+j}.\end{aligned}\tag{7.9}$$

Mathematical Tool: Geometric Series

- In going from the first to the second line of the previous slide, we noticed that:

$$\begin{aligned}\frac{v_t}{1 - \gamma_2^{-1}L^{-1}} &= (1 + \gamma_2^{-1}L^{-1} + \gamma_2^{-2}L^{-2} + \gamma_2^{-3}L^{-3} + \dots)v_t \\ &= E_t \sum_{j=0}^{\infty} \gamma_2^{-j} v_{t+j}.\end{aligned}$$

- We used the properties of the operator L^{-k} for $k \geq 1$.
- To remember the above expansion, recall that for a generic real number c with $|c| < 1$:

$$\frac{1}{1 - c} = \sum_{j=0}^{\infty} c^j.$$

- Set $c = \gamma_2^{-1}L^{-1}$ and note that $\gamma_2 > 1$, therefore the result follows.

Dynamic Solution: The Output Gap

- ▶ Using $Lx_t = x_{t-1}$, we can write (7.9) as:

$$x_t = \gamma_1 x_{t-1} - \theta \gamma_1 E_t \sum_{j=0}^{\infty} \gamma_2^{-j} v_{t+j}.$$

White Noise Assumption

- ▶ For the sake of simplicity, let us assume that v_t is a white-noise shock, therefore $E_t v_{t+j} = 0$ for each $j > 0$ and then:

$$x_t = \gamma_1 x_{t-1} - \theta \gamma_1 v_t.$$

Interpretation

- ▶ The above equation represents the dynamic solution for output gap, showing that it falls on impact when mark-up increases.

Dynamic Solution: Inflation and Interest Rates

Path of Inflation

- ▶ The path of inflation follows from (7.3) and is represented by:

$$(\pi_t - \pi) = -\theta^{-1}(\gamma_1 - 1)x_{t-1} + \gamma_1 v_t.$$

- ▶ Inflation rises above target at the time in which the markup shock hits.

Path of Interest Rate

- ▶ Using the AD equation (7.5), we derive the path consistent with optimal policy:

$$\begin{aligned}\hat{i}_t &= r_t^e + E_t(\pi_{t+1} - \pi) + \sigma^{-1}E_t(x_{t+1} - x_t) \\ \hat{i}_t &= r_t^e - (\sigma^{-1} - \theta^{-1})(1 - \gamma_1)x_t\end{aligned}\tag{7.10}$$

- ▶ This shows the nominal interest rate should follow the frictionless real rate and react inversely to the output gap (provided $\sigma < \theta$).

Interpretation of the Interest Rate Path

Not an Instrument Rule

- ▶ Equation (7.10) should **not** be interpreted as the interest rate rule the policymaker follows to implement optimal policy.
- ▶ Instead, it represents the *resulting path* of the policy rate once the central bank achieves the targeting rule (7.3).
- ▶ **Note:** If (7.10) were used as a mechanical instrument rule, it would lead to **equilibrium indeterminacy**.

The Role of R-star (r^*)

- ▶ The equation shows that matching the Wicksellian real rate is a useful check when responding to **efficient shocks**.
- ▶ However, this does **not** hold when dealing with **inefficient disturbances**.
- ▶ **Policy Implication:** Strategies focusing on R-star (frictionless rate) must be cautious, as simple tracking is insufficient during inefficient shocks.

Optimal Policy with Commitment

Inertia and Expectations

- ▶ Under optimal policy with commitment, inflation and the output gap show **inertial behavior**, since the AS equation (7.6) depends on expectations ($E_t \pi_{t+1}$). Managing these expectations is critical for maximum welfare.

Dynamic Response to a Markup Shock

- ▶ **Inflation:** Rises above target on impact, but then falls *below* target to meet it in the long run.
- ▶ **Prices:** Rise initially (more than target requires), but subsequently their growth rate falls below the target rate.
- ▶ **Interest Rate:** Inherits the persistent behavior of the output gap. Even if the shock lasts only one period, the rate remains above steady state for a long time, declining only gradually.

- 1 Introduction
- 2 Outline of the Results
- 3 Optimal policy under commitment
- 4 Optimal policy under commitment and markup shocks
- 5 Optimal Policy under Discretion**
- 6 Optimal Inflation Target
- 7 Main Takeaways

Optimal Policy Under Discretion

- ▶ **Re-optimization:** Discretion implies the policymaker re-optimizes each period; past promises are not honored.
- ▶ **Expectations:** Private agents internalize this. They form expectations knowing the policymaker will not commit to future actions.
- ▶ The forward-looking AS equation links current outcomes to expected future policy, causing discretion to yield a different inflation–output path than commitment.
- ▶ With discretion and linear-quadratic optimization problem, equilibrium allocations must be linear functions of the economy's state variables.
- ▶ In this setting, the only state variable is the markup shock v_t (assumed white noise).

Constructing the Discretionary Equilibrium

Guessed Functional Form

- ▶ Since v_t is the only state variable, we guess a linear solution of the form:

$$(\pi_t - \pi) = \pi_d v_t, \quad x_t = x_d v_t$$

Implication for Expectations

- ▶ Because v_t is white noise ($E_t v_{t+1} = 0$), inflation expectations are zero:

$$E_t(\pi_{t+1} - \pi) = E_t(\pi_d v_{t+1}) = 0$$

Optimization Problem Under Discretion

Objective

- ▶ Only the current-period loss matters, since inflation and output depend solely on the current shock v_t .

$$\min_{\pi_d, x_d} \quad \frac{1}{2}(x_d v_t)^2 + \frac{1}{2} \frac{\theta}{\kappa} (\pi_d v_t)^2$$

AS Constraint

- ▶ We substitute the guess and the expectation into the AS equation (7.6). This yields a static constraint on the coefficients:

$$\pi_d v_t = \kappa x_d v_t + v_t$$

Optimal Solution and Targeting Rule

Optimal Coefficients

- Solving the minimization problem subject to the constraint yields the optimal responses to the shock:

$$x_d = -\frac{\theta}{1 + \theta\kappa}, \quad \pi_d = \frac{1}{1 + \theta\kappa}.$$

- Combining these expressions implies a linear relationship between inflation and the output gap:

The Discretionary Targeting Rule

$$(\pi_t - \pi) + \theta^{-1}x_t = 0. \tag{7.11}$$

Comparison: Discretion vs. Commitment

Lack of Inertia

- ▶ Differently from commitment, optimal policy under discretion **does not feature any inertia**.
- ▶ It implies a different short-run response to the shock.
- ▶ Comparing the targeting rules:
 - ▶ **Discretion (7.11)**: Trade-off between inflation and the **level** of the output gap.
 - ▶ **Commitment (7.3)**: Trade-off involves the **growth rate** of the output gap ($x_t - x_{t-1}$).
- ▶ Under commitment, the policymaker cares about output growth (internalizing past levels). This reflects the necessity of fulfilling past promises, resulting in a inertial plan.

Interest Rate Rules

- ▶ We now move away from the "optimal policy" derived previously.
- ▶ Instead, we assume the policymaker follows a specific **interest-rate rule** (often called a Taylor-type rule):

$$\hat{i}_t = r_t^e + \phi_\pi(\pi_t - \pi) + \phi_x x_t. \quad (7.12)$$

- ▶ The rule incorporates a one-to-one response to the frictionless real rate, taken as perfectly predictable to isolate inefficient (markup) shocks.
- ▶ **Note:** original Taylor rule does not consider a response to the frictionless real rate.

Policy Reaction Parameters

Structure of the Rule

- ▶ Two reaction components govern policy:
 - ▶ Inflation deviation: $\phi_{\pi}(\pi_t - \pi)$
 - ▶ Output gap: $\phi_x x_t$
- ▶ Parameters satisfy $\phi_{\pi} \geq 0, \phi_x \geq 0$.

Interpretation

- ▶ ϕ_{π} : weight placed on inflation stability.
- ▶ ϕ_x : weight placed on output-gap stabilization.
- ▶ Choice of parameters encodes the central bank's preference mix under interest rule-based policy.

Model Under an Interest Rate Rule

System Reduction

- ▶ Adding (7.12) to (7.5) and (7.6) initially gives three equations for three variables ($\hat{i}_t$, π_t , x_t).
- ▶ We use (7.6) and (7.12) into (7.5) to substitute for $\hat{i}_t$ and $E_t(\pi_{t+1} - \pi)$.
- ▶ This compresses the model into a system of two stochastic linear difference equations:

$$E_t \begin{bmatrix} \pi_{t+1} - \pi \\ x_{t+1} \end{bmatrix} = \begin{bmatrix} \frac{1}{\beta} & -\frac{\kappa}{\beta} \\ \sigma(\phi_\pi - \frac{1}{\beta}) & 1 + \sigma\phi_x + \sigma\frac{\kappa}{\beta} \end{bmatrix} \begin{bmatrix} \pi_t - \pi \\ x_t \end{bmatrix} + \begin{bmatrix} -\frac{1}{\beta} \\ \frac{\sigma}{\beta} \end{bmatrix} v_t \quad (7.13)$$

Implication

- ▶ Because the rule offsets movements in the frictionless real rate, dynamics of inflation and the output gap depend solely on the markup disturbance.

Determinacy and Stability Analysis

1. Blanchard-Kahn Conditions

- ▶ **Rule:** The number of eigenvalues within the unit circle must equal the number of predetermined variables.
- ▶ **Application:** In system (7.13), there are **no predetermined variables** (no terms indexed $t - 1$).
- ▶ **Requirement:** For a unique, bounded solution, both eigenvalues must lie outside the unit circle ($|\gamma| > 1$).

Matrix $\mathcal{V}$

2. The Characteristic Polynomial

- Consider the matrix $\mathcal{V}$ from system (7.13):

$$\mathcal{V} = \begin{bmatrix} \frac{1}{\beta} & -\frac{\kappa}{\beta} \\ \sigma(\phi_\pi - \frac{1}{\beta}) & 1 + \sigma\phi_x + \sigma\frac{\kappa}{\beta} \end{bmatrix}$$

- The characteristic polynomial is given by $P(\gamma) = \gamma^2 - \text{tr}[\mathcal{V}]\gamma + \det[\mathcal{V}]$.
- Trace and determinant are both positive:

$$\begin{aligned} \text{tr}[\mathcal{V}] &= \left(1 + \frac{1}{\beta} + \sigma\phi_x + \sigma\frac{\kappa}{\beta}\right) > 0 \\ \det[\mathcal{V}] &= \frac{1}{\beta} (1 + \sigma\phi_x + \sigma\kappa\phi_\pi) > 0 \end{aligned}$$

Determinacy under Interest Rate Rules

Location of the Eigenvalues

- ▶ Since $\text{tr}[\mathcal{V}] > 0$ and $\det[\mathcal{V}] > 0$, both eigenvalues of the system matrix $\mathcal{V}$ are positive.
- ▶ To determine whether the roots lie inside or outside the unit circle, we evaluate the characteristic polynomial $P(\gamma)$ at key points:
 - ▶ $P(0) = \det[\mathcal{V}] > 0$
 - ▶ $P(\infty) = \infty$
- ▶ **Determinacy condition:** For both eigenvalues to lie outside the unit circle (no predetermined variables), we require:

$$P(1) = \sigma\phi_x(\beta^{-1} - 1) + \sigma\kappa\beta^{-1}(\phi_\pi - 1) > 0. \quad (7.14)$$

The Taylor Principle

The Taylor Principle

- ▶ Condition (7.14) is known as the **generalized Taylor principle**.
- ▶ It ensures a **unique and stable equilibrium** by preventing self-fulfilling fluctuations driven by expectations.

Special Case

- ▶ If the policy rule does not respond to the output gap ($\phi_x = 0$), the condition simplifies to:

$$\phi_\pi > 1.$$

- ▶ This is the classic **Taylor principle**.

Economic Intuition

- ▶ A one-percent increase in inflation must trigger a *more than one-percent* increase in the nominal interest rate.
- ▶ This raises the **real interest rate**, dampening aggregate demand and stabilizing inflation.

Characterizing the Unique Solution

Prerequisites

- ▶ Under condition (7.14), we can now characterize the unique and stable solution.
- ▶ **Assumption:** For simplicity, let us assume again that v_t follows a white-noise process.

Method of Undetermined Coefficients

- ▶ One way to characterize the solution is to guess that the equilibrium variables are linear functions of the shock v_t .
- ▶ We posit the following form:

$$\pi_t - \pi = \pi_r v_t,$$

$$x_t = x_r v_t.$$

Deriving the Parameter Restrictions

- ▶ Inserting the hypothesized solution into the first line of system (7.13):

$$\pi_r E_t v_{t+1} = \frac{1}{\beta} \pi_r v_t - \frac{\kappa}{\beta} x_r v_t - \frac{1}{\beta} v_t.$$

- ▶ Since $E_t v_{t+1} = 0$ (white noise), we simplify to obtain the first restriction between unknown parameters:

$$\pi_r = \kappa x_r + 1. \quad (7.15)$$

- ▶ From the second line of (7.13), we obtain:

$$x_r E_t v_{t+1} = \sigma \left(\phi_\pi - \frac{1}{\beta} \right) \pi_r v_t + \left(1 + \sigma \phi_x + \sigma \frac{\kappa}{\beta} \right) x_r v_t + \frac{\sigma}{\beta} v_t.$$

- ▶ Setting the LHS to zero and rearranging yields the second restriction:

$$\left(\frac{1}{\beta} - \phi_\pi \right) \pi_r = \left(\sigma^{-1} + \phi_x + \frac{\kappa}{\beta} \right) x_r + \frac{1}{\beta}. \quad (7.16)$$

Solution and Policy Trade-offs

- Combining the restrictions (7.15) and (7.16), we obtain the final coefficients:

$$x_r = -\frac{\phi_\pi}{\sigma^{-1} + \phi_x + \kappa\phi_\pi}, \quad \pi_r = \frac{\sigma^{-1} + \phi_x}{\sigma^{-1} + \phi_x + \kappa\phi_\pi}$$

General Dynamics

- The parameters satisfy $\pi_r \geq 0$ and $x_r \leq 0$.
- A markup shock puts upward pressure on inflation and reduces the output gap (relative to the efficient level).

Limiting cases under Interest Rate Rule

Varying policy parameters ϕ_π and ϕ_x reveals the trade-off along the structural constraint $\pi_r = \kappa x_r + 1$.

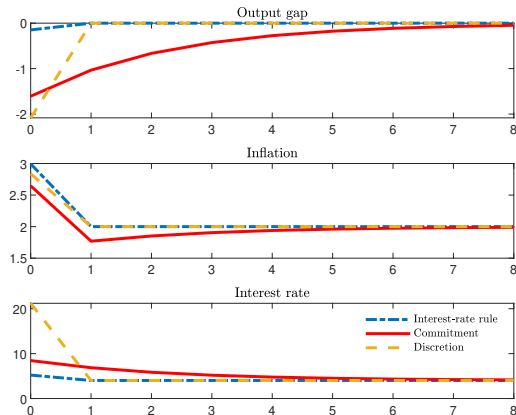
Strict Inflation Targeting ($\phi_\pi \rightarrow \infty$)

- ▶ **Result:** $\pi_r \rightarrow 0$ and $x_r \rightarrow -1/\kappa$.
- ▶ Inflation is perfectly stabilized.
- ▶ The entire shock is absorbed by a sharp fall in output.

Strict Output Targeting ($\phi_x \rightarrow \infty$)

- ▶ **Result:** $x_r \rightarrow 0$ and $\pi_r \rightarrow 1$.
- ▶ Output is perfectly stabilized.
- ▶ There is full pass-through of the markup shock into inflation.

Dynamic Responses after a Markup Shock

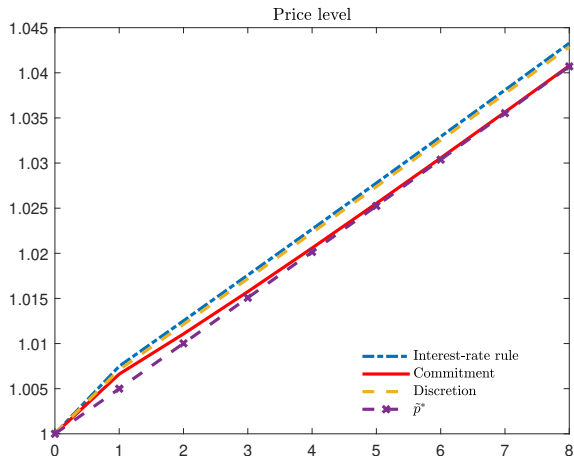


Note: The Figure uses the following parametrization: $\beta = 0.995$, $\kappa = 0.02$, $\sigma = 0.5$, $\theta = 10$, $\eta = 0.47$, $\phi_y = 0.5$, $\phi_\pi = 1.5$. The one-period markup shock (v) is set at 0.25%. The model is calibrated quarterly. Inflation target π is set at a 2% annual rate. Output gap in %, inflation and interest rate in annualized %.

Comparison of Policies

- ▶ We compare the impulse responses of the output gap, inflation, and interest rate after a positive temporary markup shock ($v = 0.25\%$) under commitment, discretion, and an interest-rate rule.
- ▶ **Commitment:** Variables display a **inertial response**, despite the shock lasting only one period.
- ▶ **Discretion & Rule:** The response dies out immediately as the shock reverts to zero.

Dynamic Response of the Price Level



Note: Price path following the shock. The line $\tilde{p}^*$ represents the 2% trend.

Comparison of Policies

- ▶ **Commitment:** The initial rise in prices is **undone** in subsequent periods. The price level returns to the pre-shock trend ($\tilde{p}^*$).
- ▶ **Discretion & Rule:** Prices jump and then grow at the 2% rate from the *new, higher* level. They never reconnect to the initial trend.

Conclusion

- ▶ Under discretion, "bygones are bygones."
- ▶ Commitment maintains long-term price stability.

- 1 Introduction
- 2 Outline of the Results
- 3 Optimal policy under commitment
- 4 Optimal policy under commitment and markup shocks
- 5 Optimal Policy under Discretion
- 6 Optimal Inflation Target**
- 7 Main Takeaways

Why a Positive Inflation Target? ($\pi > 0$)

Three main reasons justify a positive inflation target:

- 1 **ZLB Buffer:** A higher inflation target implies a higher steady-state nominal interest rate. This provides a safety margin for the central bank to offset negative shocks to the *frictionless real rate* $\hat{r}_t^e$ before the zero lower bound ($i_t \geq 0$) becomes binding.
- 2 **Downward Wage Rigidity:** Nominal wages are rigid downward. Positive inflation acts as “grease” in the labor market (Tobin 1972), allowing real wages to adjust downward without nominal wage cuts, thereby reducing employment and output volatility.
- 3 **Debt–Deflation Risk:** With $\pi = 0$, adverse shocks may trigger deflation, increasing the real burden of nominal debt (Fisher 1933). This can induce a self-reinforcing contraction in demand, output, and prices.

Quantitative Targets: Research Findings

While qualitative arguments support $\pi > 0$, quantitative research narrows the optimal range based on specific structural features:

- ▶ **ZLB and Indexing:** Accounting for ZLB costs suggests $\pi^* \approx 2\%$ (Coibion, Gorodnichenko, and Wieland 2012). Note: If firms do not index to a target, costs of $\pi > 0$ rise, potentially underestimating the optimal rate.
- ▶ **Endogenous Entry:** In models with product variety and firm entry, deviations from long-run price stability are optimal (Bilbiie, Fujiwara, and Gironi 2014).
- ▶ **Productivity Trends:** In frameworks with heterogeneous productivity trends across firms, the optimal rate is estimated between 1% and 3% (Adam and Weber 2019).
- ▶ **Measurement Bias:** A positive target acknowledges quality improvements and new goods not previously captured in standard price indices.

The Upper Bound: Costs of High Inflation

Why not target much higher inflation? Significant distortions arise as π increases:

- ▶ **Tax Distortions:** Interaction with non-indexed tax brackets increases effective tax rates and distorts incentives (Feldstein 1997).
- ▶ **Information and Signal Noise:** High inflation obscures relative price signals and complicates long-term planning for investment and credit.
- ▶ **Volatility and Misallocation:** High π is empirically associated with higher inflation volatility (International Monetary Fund 2015), leading to resource misallocation.
- ▶ **Expectations De-anchoring:** High, volatile inflation increases the "attention" agents pay to prices, making expectations more volatile and harder to control.

Conclusion: Current targets (1%–3%) represent a reasonable balance between avoiding the ZLB and minimizing distortionary costs.

- 1 Introduction
- 2 Outline of the Results
- 3 Optimal policy under commitment
- 4 Optimal policy under commitment and markup shocks
- 5 Optimal Policy under Discretion
- 6 Optimal Inflation Target
- 7 Main Takeaways**

Main Takeaways

- ▶ **Flexible Inflation Targeting:** optimal under commitment; allows temporary inflation deviations only in exchange for deviations of **output growth from the efficient growth trend**.
- ▶ **Optimal Targeting Rule:**

$$(\pi_t - \pi) + \theta^{-1}(\Delta \hat{Y}_t - \Delta \hat{Y}_t^e) = 0.$$

- ▶ **History Dependence:** commitment is equivalent to **price-level targeting**; it implies **price-level reversion** after inefficient shocks.
- ▶ **Discretion:** sub-optimal because it trades off inflation with the *level* of the output gap and lacks inertia (expectations management).
- ▶ **Taylor Rule:** determinacy requires a generalized Taylor principle; when $\phi_x = 0$ it reduces to $\phi_\pi > 1$.
- ▶ **The Numerical Target:** structural frictions (ZLB, wage rigidity, debt-deflation) justify a positive π and support targets in the **1–3% range**.

8 Appendix: Derivations

9 Bibliography

Derivation of equation (7.1)

Consider the expected utility of the consumers (5.1) and write it as

$$U_{t_0} = E_{t_0} \left\{ \sum_{t=t_0}^{+\infty} \beta^{t-t_0} \xi_t [(U(C_t) - H(L_t)\Delta_t)] \right\},$$

$$U(C_t) = \frac{C_t^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}$$

$$H(L_t) = \frac{L_t^{1+\eta}}{1+\eta},$$

$$\Delta_t \equiv \int_0^1 \left(\frac{p_t(j)}{P_t} \right)^{-\theta(1+\eta)} dj.$$

Observe, indeed, that

$$L_t(j) = \frac{Y_t(j)}{A_t} = \left(\frac{p_t(j)}{P_t} \right)^{-\theta} \frac{Y_t}{A_t} = \left(\frac{p_t(j)}{P_t} \right)^{-\theta} L_t.$$

Note the following second-order Taylor approximation

$$\begin{aligned}
 \frac{C_t^{1-\frac{1}{\tilde{\sigma}}}}{1-\frac{1}{\tilde{\sigma}}} &= \frac{C^{1-\frac{1}{\tilde{\sigma}}}}{1-\frac{1}{\tilde{\sigma}}} + C^{-\frac{1}{\tilde{\sigma}}}(C_t - C) - \frac{1}{2} \frac{1}{\tilde{\sigma}} C^{-\frac{1}{\tilde{\sigma}}-1} (C_t - C)^2 + \mathcal{O}(\|\xi\|^3) \\
 &= \frac{C^{1-\frac{1}{\tilde{\sigma}}}}{1-\frac{1}{\tilde{\sigma}}} + C^{1-\frac{1}{\tilde{\sigma}}} \hat{C}_t + \frac{1}{2} \left(1 - \frac{1}{\tilde{\sigma}}\right) C^{1-\frac{1}{\tilde{\sigma}}} \hat{C}_t^2 + \mathcal{O}(\|\xi\|^3)
 \end{aligned} \tag{7.17}$$

where $\mathcal{O}(\|\xi\|^3)$ collects terms of order higher than the second and where, in the second line, we have used the following approximation:

$$\left(\frac{C_t - C}{C} \right) = \hat{C}_t + \frac{1}{2} \hat{C}_t^2 + \mathcal{O}(\|\xi\|^3).$$

Similarly, we can write

$$\begin{aligned}
H(L_t)\Delta_t &= H(L) + H_l(L_t - L) + \frac{1}{2}H_{ll}(L_t - L)^2 + \\
&\quad + H(L)(\Delta_t - 1) + \mathcal{O}(\|\xi\|^3) \\
&= H(L) + H_l L \hat{L}_t + \frac{1}{2}H_l L (1 + \eta) \hat{L}_t^2 + \\
&\quad + H(L)(\Delta_t - 1) + \mathcal{O}(\|\xi\|^3),
\end{aligned} \tag{7.18}$$

having used $\eta = H_{ll}Y/H_l$ and the fact that the expansion of Δ_t is of second-order magnitude, as it will be shown.

Combining (7.17) and (7.18), we obtain

$$\begin{aligned}
U(C_t) - H(L_t)\Delta_t &= U(C) - H(L)\Delta + U_c \hat{C}_t + \frac{1}{2}U_c C \left(1 - \frac{1}{\tilde{\sigma}}\right) \hat{C}_t^2 - H_l L \hat{L}_t + \\
&\quad - \frac{1}{2}H_l L (1 + \eta) \hat{L}_t^2 - H(Y)(\Delta_t - 1) + \mathcal{O}(\|\xi\|^3).
\end{aligned} \tag{7.19}$$

Note that, given $Y_t = A_t L_t$, it holds exactly that

$$\hat{L}_t = \hat{Y}_t - \hat{A}_t, \quad (7.20)$$

whereas a second-order expansion of the equilibrium in the goods market $Y_t = C_t + G_t$ implies

$$Y_t - Y = C_t - C + G_t - G$$

and therefore

$$\hat{Y}_t + \frac{1}{2} \hat{Y}_t^2 = \frac{C}{Y} \left(\hat{C}_t + \frac{1}{2} \hat{C}_t^2 \right) + \hat{G}_t + \mathcal{O}(\|\xi\|^3)$$

using definition $\hat{G}_t = (G_t - G)/Y$. We can then write

$$\hat{C}_t = \frac{Y}{C} \left(\hat{Y}_t - \hat{G}_t \right) + \frac{1}{2} \frac{Y}{C} \hat{Y}_t^2 - \frac{1}{2} \hat{C}_t^2 + \mathcal{O}(\|\xi\|^3)$$

and therefore

$$\hat{C}_t = \frac{Y}{C} \left(\hat{Y}_t - \hat{G}_t \right) + \frac{1}{2} \frac{Y}{C} \hat{Y}_t^2 - \frac{1}{2} \frac{Y^2}{C^2} \left(\hat{Y}_t - \hat{G}_t \right)^2 + \mathcal{O}(\|\xi\|^3). \quad (7.21)$$

We can combine (7.20) and (7.21) in (7.19) to obtain

$$\begin{aligned}
U(C_t) - H(L_t)\Delta_t &= C^{1-\frac{1}{\tilde{\sigma}}} \left[\frac{Y}{C} (\hat{Y}_t - \hat{G}_t) + \frac{1}{2} \frac{Y}{C} \hat{Y}_t^2 - \frac{1}{2} \frac{Y^2}{C^2} (\hat{Y}_t - \hat{G}_t)^2 \right] \\
&\quad - L^{1+\eta} (\hat{Y}_t - \hat{A}_t) + \frac{1}{2} \left(1 - \frac{1}{\tilde{\sigma}} \right) C^{1-\frac{1}{\tilde{\sigma}}} \frac{Y^2}{C^2} (\hat{Y}_t - \hat{G}_t)^2 \\
&\quad - \frac{1}{2} (1 + \eta) L^{1+\eta} (\hat{Y}_t - \hat{A}_t)^2 - H(Y)(\Delta_t - 1) + \mathcal{O}(\|\xi\|^3),
\end{aligned}$$

by neglecting constant terms. Note that in the steady state $C^{-\frac{1}{\tilde{\sigma}}}/L^\eta = 1/A = L/Y$, therefore $C^{-\frac{1}{\tilde{\sigma}}}Y = L^{1+\eta}$. We can then simplify the above equation by also neglecting all terms independent of policy to

$$\begin{aligned}
U(C_t) - \Delta_t H(L_t) &= L^{1+\eta} \left\{ -\frac{\hat{\Delta}_t}{(1+\eta)} + \frac{1}{2} \hat{Y}_t^2 - \frac{1}{2} \frac{1}{\tilde{\sigma}} \frac{Y}{C} (\hat{Y}_t - \hat{G}_t)^2 \right. \\
&\quad \left. - \frac{1}{2} (1 + \eta) (\hat{Y}_t - \hat{A}_t)^2 \right\} + \mathcal{O}(\|\xi\|^3).
\end{aligned}$$

Use now definition $\sigma \equiv \tilde{\sigma}C/Y$ and that of efficient level of output

$$\hat{Y}_t^e = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_t + \frac{\sigma^{-1}}{\sigma^{-1} + \eta} \hat{G}_t$$

to write

$$U(C_t) - \Delta_t H(L_t) = L^{1+\eta} \left\{ -\frac{\hat{\Delta}_t}{(1 + \eta)} - \frac{\sigma^{-1} + \eta}{2} (\hat{Y}_t - \hat{Y}_t^e)^2 \right\} + \mathcal{O}(\|\xi\|^3). \quad (7.22)$$

Consider now the following approximation

$$\begin{aligned} \xi_t [U(C_t) - \Delta_t H(L_t)] &= (\xi_t - \xi) [U(C) - \Delta H(L)] + \xi [U(C_t) - \Delta_t H(L_t)]^{2^{nd}} + \\ &\quad (\xi_t - \xi) [U_c(C_t - C) - H_l(L_t - L)] + \mathcal{O}(\|\xi\|^3) \end{aligned}$$

in which the second addendum on the right-hand side of the first line is meant to represent the second-order approximation found above. Given the steady-state assumptions made above, it follows that

$$\xi_t[U(C_t) - \Delta_t H(L_t)] = [U(C_t) - \Delta_t H(L_t)]^{2^{nd}} + \text{t.i.p.} + O(\|\xi\|^3),$$

in which t.i.p. represents terms independent of policy.

Therefore, using (7.22), we can write a second-order approximation of the expected discounted utility as

$$U_{t_0} = -L^{1+\eta} E_{t_0} \left\{ \sum_{t=t_0}^{+\infty} \beta^{t-t_0} \left[-\frac{\hat{\Delta}_t}{(1+\eta)} - \frac{\sigma^{-1} + \eta}{2} (\hat{Y}_t - \hat{Y}_t^e)^2 \right] \right\} + \text{t.i.p.} + O(\|\xi\|^3). \quad (7.23)$$

Recall the definition of Δ_t

$$\Delta_t \equiv \int_0^1 \left(\frac{p_t(j)}{P_t} \right)^{-\theta(1+\eta)} dj$$

which can be written recursively as

$$\Delta_t = \alpha \Delta_{t-1} \left(\frac{\Pi_t}{\Pi} \right)^{\theta(1+\eta)} + (1 - \alpha) \left(\frac{\mathcal{P}_t^*}{P_t} \right)^{-\theta(1+\eta)}.$$

Using (5.22) to substitute $\mathcal{P}_t^*/P_t$, we obtain

$$\Delta_t = \alpha \Delta_{t-1} \left(\frac{\Pi_t}{\Pi} \right)^{\theta(1+\eta)} + (1 - \alpha) \left(\frac{1 - \alpha \left(\frac{\Pi_t}{\Pi} \right)^{\theta-1}}{1 - \alpha} \right)^{\frac{\theta(1+\eta)}{\theta-1}}. \quad (7.24)$$

Take a second-order Taylor expansion around the steady state in which $\Delta_t = 1$ and $\Pi_t = \Pi$ to obtain

$$\begin{aligned}
 \Delta_t - 1 &= \alpha(\Delta_{t-1} - 1) + \alpha\theta(1 + \eta) \left(\frac{\Pi_t - \Pi}{\Pi} \right) + \\
 &\quad + \frac{1}{2} \alpha\theta(1 + \eta)(\theta(1 + \eta) - 1) \left(\frac{\Pi_t - \Pi}{\Pi} \right)^2 + \\
 &\quad - \alpha\theta(1 + \eta) \left(\frac{\Pi_t - \Pi}{\Pi} \right) + \\
 &\quad + \frac{1}{2} \frac{\alpha\theta(1 + \eta)}{(\alpha - 1)} (\theta + \alpha - \theta(1 + \eta)\alpha - 2) \left(\frac{\Pi_t - \Pi}{\Pi} \right)^2 + \\
 &\quad + \mathcal{O}(\|\xi\|^3)
 \end{aligned}$$

which can be written as

$$\begin{aligned}\hat{\Delta}_t &= \alpha \hat{\Delta}_{t-1} + \frac{1}{2} \alpha \theta (1 + \eta) (\theta (1 + \eta) - 1) (\pi_t - \pi)^2 + \\ &\quad \frac{1}{2} \frac{\alpha \theta (1 + \eta)}{(\alpha - 1)} (\theta + \alpha - \theta (1 + \eta) \alpha - 2) (\pi_t - \pi)^2 + \\ &\quad + \mathcal{O}(\|\xi\|^3)\end{aligned}$$

having used

$$\Pi_t = \Pi \left[1 + (\pi_t - \pi) + \frac{1}{2} (\pi_t - \pi)^2 \right] + \mathcal{O}(\|\xi\|^3).$$

We can simplify the above equation to

$$\hat{\Delta}_t = \alpha \hat{\Delta}_{t-1} + \frac{\alpha}{1 - \alpha} \theta (1 + \eta) (1 + \eta \theta) \frac{(\pi_t - \pi)^2}{2} + \text{t.i.p.} + \mathcal{O}(\|\xi\|^3).$$

Now note that

$$\hat{\Delta}_t = \alpha^{t-t_0+1} \hat{\Delta}_{t_0-1} + \frac{1}{2} \frac{\alpha\theta}{(1-\alpha)} (1+\eta)(1+\eta\theta) \sum_{s=t_0}^t \alpha^{t-s} (\pi_s - \pi)^2 + \text{t.i.p.} + O(\|\xi\|^3)$$

and therefore

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \hat{\Delta}_t = \frac{1}{2} \frac{\alpha\theta(1+\eta)(1+\eta\theta)}{(1-\alpha)(1-\alpha\beta)} \sum_{t=t_0}^{\infty} \beta^{t-t_0} (\pi_t - \pi)^2 + \text{t.i.p.} + O(\|\xi\|^3), \quad (7.25)$$

neglecting initial condition $\hat{\Delta}_{t_0-1}$.

Combining and inserting this result into the expected discounted value of the approximated utility flow, (7.23), we finally obtain

$$U_{t_0} = -(\sigma^{-1} + \eta)L^{1+\eta}E_{t_0} \left\{ \sum_{t=t_0}^{+\infty} \beta^{t-t_0} \left[\frac{1}{2}(\hat{Y}_t - \hat{Y}_t^*)^2 + \frac{1}{2} \frac{\theta}{\kappa} (\pi_t - \pi)^2 \right] \right\} \\ + \text{t.i.p.} + O(\|\xi\|^3),$$

from which loss function (7.1) follows.

[Back to Loss Function](#)

Conditions for determinacy

This Appendix discusses the conditions under which a system of stochastic linear difference equations has a unique and stable solution. Blanchard-Kahn conditions require that the number of predetermined variables of the system be equal to the number of eigenvalues within the unit circle.¹ We provide a very informal reasoning for this result by discussing a simple example of linear stochastic difference equations. There are four cases to consider.

¹See Blanchard and Kahn 1980.

Case I

Consider a linear stochastic difference equation in the variable x_t of the form

$$E_t x_{t+1} = \lambda x_t \quad (7.26)$$

where λ , which is the eigenvalue, is outside the unit circle, i.e. $|\lambda| \geq 1$. The solutions of (7.26) are of the form

$$x_t = c \xi_t \lambda^t \quad (7.27)$$

for some $c \in \mathcal{R}$, with ξ_t being a martingale process, i.e. $E_t \xi_{t+1} = \xi_t$. Indeed, it can be verified that solutions (7.27) satisfy the difference equation (7.26) by inserting it into (7.26). There is no predetermined variable in equation (7.26), meaning that there is no initial condition for variable x_t . Since the eigenvalue is outside the unit circle, the Blanchard-Kahn conditions are satisfied. There is then a unique and bounded solution. Indeed, just set $c = 0$ into (7.27). Solution $x_t = 0$ is the only one which is stable, since the others diverge, given that $|\lambda| \geq 1$.

Case II

Consider again the same stochastic linear difference equation (7.26), but now $|\lambda| < 1$. The solutions are again of the form (7.27). However, $x_t = 0$ is no longer a unique and stable solution since even the solutions with $c \neq 0$ converge to it. Therefore, when the number of eigenvalues within the unit circle is higher than the number of predetermined variables, there are multiple stable solutions.

Case III

Consider a linear stochastic difference equation in the variable x_t of the form

$$x_t = \lambda x_{t-1} + \varepsilon_t \quad (7.28)$$

where λ , which is the eigenvalue, is inside the unit circle, i.e. $|\lambda| < 1$. There is one predetermined variable and one eigenvalue within the unit circle. The Blanchard-Kahn conditions are satisfied. Indeed, (7.28) is already the stable and unique solution.

Case IV

Consider again linear stochastic difference equation (7.28), but now with $|\lambda| \geq 1$. The number of eigenvalues inside the unit circle is lower than the number of predetermined variables. In this case there is no stable solution. Indeed, solution (7.28) is now unstable.

[Back to Difference Equation](#)

8 Appendix: Derivations

9 Bibliography

References I

-  Adam, Klaus and Henning Weber (2019). “Optimal Trend Inflation”. In: *American Economic Review* 109.2, pp. 702–37.
-  Bilbiie, Florin, Ippei Fujiwara, and Fabio Ghironi (2014). “Optimal Monetary Policy with Endogenous Entry and Product Variety”. In: *Journal of Monetary Economics* 64, pp. 162–196.
-  Blanchard, Olivier and Charles Kahn (1980). “The Solution of Linear Difference Models under Rational Expectations”. In: *Econometrica* 48, pp. 1305–1311.
-  Coibion, Olivier, Yuriy Gorodnichenko, and Johannes Wieland (2012). “The Optimal Inflation Rate in New Keynesian Models: Should it be Higher?” In: *Review of Economic Studies* 79.4, pp. 1371–1406.
-  Feldstein, Martin (1997). “The Costs and Benefits of Going from Low Inflation to Price Stability”. In: *Reducing Inflation: Motivation and Strategy*. Ed. by Christina D. Romer and David H. Romer. University of Chicago Press, pp. 123–156.

References II

-  Fisher, Irving (1933). “The Debt-Deflation Theory of Great Depressions”. In: *Econometrica* 1.4, pp. 337–357.
-  International Monetary Fund (Sept. 2015). *World Economic Outlook*. Tech. rep. IMF Staff Report.
-  Tobin, James (1972). “Inflation and Unemployment”. In: *The American Economic Review* 62.1, pp. 1–18.